EXERCISE 11.12

Integrate the following functions with respect to x :

(1) (i) $\sqrt{x^2+2x+10}$ (ii) $\sqrt{x^2-2x-3}$ (iii) $\sqrt{(6-x)(x-4)}$

(2) (i) $\sqrt{9-(2x+5)^2}$ (ii) $\sqrt{81+(2x+1)^2}$ (iii) $\sqrt{(x+1)^2-4}$

EXERCISE 11.13

Choose the correct or the most suitable answer from given four alternatives.

(1) If $\int f(x)dx = g(x)+c$, then $\int f(x)g'(x)dx$

(1) $\int (f(x))^2 dx$ (2) $\int f(x)g(x)dx$ (3) $\int f'(x)g(x)dx$ (4) $\int (g(x))^2 dx$

(2) If $\int \frac{3^{\frac{1}{x}}}{x^2}dx = k\ (3^{\frac{1}{x}})+c$, then the value of k is

(1) $\log 3$ (2) $-\log 3$ (3) $-\frac{1}{\log 3}$ (4) $\frac{1}{\log 3}$

(3) If $\int f'(x)e^{x^2}dx = (x-1)e^{x^2}+c$, then $f(x)$ is

(1) $2x^3-\frac{x^2}{2}+x+c$ (2) $\frac{x^3}{2}+3x^2+4x+c$ (3) x^3+4x^2+6x+c (4) $\frac{2x^3}{3}-x^2+x+c$

(4) The gradient (slope) of a curve at any point (x, y) is $\frac{x^2-4}{x^2}$. If the curve passes through the point (2, 7), then the equation of the curve is

(1) $y = x+\frac{4}{x}+3$ (2) $y = x+\frac{4}{x}+4$ (3) $y = x^2+3x+4$ (4) $y = x^2-3x+6$

(5) $\int \frac{e^x(1+x)}{\cos^2(xe^x)}\ dx$ is

(1) $\cot(xe^x)+c$ (2) $\sec(xe^x)+c$ (3) $\tan(xe^x)+c$ (4) $\cos(xe^x)+c$

(6) $\int \frac{\sqrt{\tan x}}{\sin 2x}dx$ is

(1) $\sqrt{\tan x}+c$ (2) $2\sqrt{\tan x}+c$ (3) $\frac{1}{2}\sqrt{\tan x}+c$ (4) $\frac{1}{4}\sqrt{\tan x}+c$

(7) $\int \sin^3 x dx$ is

(1) $\frac{-3}{4}\cos x-\frac{\cos 3x}{12}+c$ (2) $\frac{3}{4}\cos x+\frac{\cos 3x}{12}+c$

(3) $\frac{-3}{4}\cos x+\frac{\cos 3x}{12}+c$ (4) $\frac{-3}{4}\sin x-\frac{\sin 3x}{12}+c$

(8) $\int \frac{e^{6\log x}-e^{5\log x}}{e^{4\log x}-e^{3\log x}}dx$ is

(1) $x+c$ (2) $\frac{x^3}{3}+c$ (3) $\frac{3}{x^3}+c$ (4) $\frac{1}{x^2}+c$

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